MRO Reverse Engineering Study

Assignment #4

Team 13

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Contents

1 Attitude and Orbit Control Subsystem

The Attitude and Orbit Control Subsystem (AOCS) provides the hardware and the software capabilities to perform attitude measurements, estimation, guidance and control required across all of the control modes of the mission. The AOCS allows to perform corrective maneuvers on the trajectory to Mars, while keeping the solar arrays and the HGA pointed toward the Sun and Earth respectively to satisfy the science requirements. This subsystem relies on two star trackers, sixteen Sun sensors, and two inertial measurement units to determine the Mars Reconnaissance Orbiter (MRO) attitude. A set of four reaction wheels (RW) and eight Reaction Control System (RCS) thrusters are used to control the attitude [7, 8].

1.1 Pointing Performance Related Requirements

The design of AOCS was driven by two main performance requirements: target relative pointing and pointing stability. The first defines how well a body-fixed vector must point at a specified target on the surface of Mars, ensuring that it is captured in the instrument Field-of-view (FOV). The second specifies how still the spacecraft must remain over a specified period of time, ensuring that the image quality meets the science objective [5]. The stability requirements for each imaging instrument are summarized in Table 1, along with the time interval in which this accuracy shall be maintained. It can be seen that the HiRISE requirements are, by far, the most challenging with respect to the other instruments. This is also valid for target pointing requirements, for which the HiRISE boresight shall point to a target on Mars within 0.7 mrad (3-sigma) in the FOV cross-track direction and 2 mrad in down-track direction.

Instrument	**Magnitude**	**Window**
HiRISE	0.0032 mrad	12 msec
CTX	0.020 mrad	2.5 msec
CRISM	0.032 mrad	200 msec
ONC	0.033 mrad	300 msec
MCS	3 mrad	16 sec

Table 1: MRO Science instrument pointing stability requirements [5]

A crucial aspect to consider in the design of the AOCS is the simultaneous operation of multiple instruments, which makes it necessary to account for and correct (as much as possible) the instrument misalignment [5]. The pointing requirements of the Tracking, Telemetry and Telecommand Subsystem (TTMTC) and the power subsystem are mainly handled by the gimbal mechanisms of the High-Gain Antenna (HGA) and the solar arrays respectively, and thus are not drivers for the AOCS design.

1.2 Architecture and Rationale

Even if the most challenging requirements derive from HiRISE, the architecture design must take into account all the instruments used for science objectives, as well as the conditions for proper functioning of the spacecraft. MRO is designed to support continuous nadir-oriented operations at Mars and periodic maneuvering to obtain high resolution images of targets. The orbiter is able to roll and capture data ±30° cross-track of nadir, which is operationally held below 10° as much as possible to meet scientific requirements. MRO is equipped with strongly reliable sensors and actuators, always granting redundancy in case of failure. In the following paragraphs the different components of AOCS are specifically described.

1.2.1 Sensors

MRO is equipped with two Honeywell Miniature Inertial Measurement Units (MIMUs), two Galileo Avionica A-STR star trackers and sixteen Sun sensors.

MIMU: each device is a combination of three accelerometers and three ring laser gyroscopes (one gyro and one accelerometer for each axis). The accelerometers measure the accelerations to provide information about the firings of the thrusters, while the gyroscopes measure the rotational velocity of the spacecraft. MIMUs allow also to estimate the orbiter orientation for brief periods when the angular rate is too high for star trackers [8]. These devices are highly reliable, ensuring less than 2200 Failure in Time (FIT), meaning that it can have a maximum of 2200 failures over a period of one billion hours [2]. The second MIMU is equipped for backup purposes in case of failure of the primary device.
Sun Sensors: they are distributed on the solar arrays and orbiter bus, allowing the spacecraft to locate the Sun. Only eight of the sixteen sensors are used, while the others are equipped for redundancy. They can only provide information about the presence of the Sun in their FOV. Combining the information of all the sensors, the on-board computer can roughly estimate the position of the Sun. These devices are typically used during the waking-up of the spacecraft after launch or safe mode, giving the spacecraft enough information to re-orient towards and get power from the Sun. However, information from Sun sensors alone is not sufficient to estimate precise position and attitude [8].
Star Trackers: the star trackers provide full knowledge of the orbiter orientation, identifying the stars around and matching them with its database. Due to the crucial role of this device, a second redundant star tracker is included, in case of failure. This device continuously takes digital images of the surrounding space and compares them with an on-board catalog until it can identify the stars in the images [5]. The autonomous A-STR star tracker is strongly reliable (TRL-9) and offers high performances with its CCD detector. It is a medium FOV tracker (16.4° x 16.4°), able to identify up to 10 stars with a tracking rate of 2°/s. Despite its reduced dimensions (195 x 175 x 290.5 mm), this device is affected by very low random errors, i.e. Noise Equivalent Error (NEA), of less than 25 arcsec for pitch and yaw, and less than 230 arcsec for roll [6].

1.2.2 Actuators

The actuators allow to translate attitude commands into actual control motion of the orbiter. The MRO is equipped with four Honeywell Constellation Series HR16 RWs and eight small Aerojet MR-103D thrusters, comprising the RCS.

RW: this assembly is composed of three wheels mounted perpendicularly to each other (one for each rotational axis, commonly aligned along the body-frame axis), with a fourth wheel mounted in a skewed direction as a redundant unit in the event of failure of one of the primary wheels. RWs are used during all the mission phases [4]. They provide slow and steady turns, which are suitable for high-resolution images required for science objectives. Each wheel has a mass of about 12 kg and a maximum angular velocity of 6000 rpm. The torque produced can be regulated controlling the wheels with DC brushless electric motors, ensuring low jitter disturbance and reaching a maximum reaction torque of 0.4 Nm. The power consumption of each unit requires at their peak 195 W when applying torque, and about 22 W in their steady state at 6000 rpm.
RCS: this system relies on eight thrusters, coupled in pairs, that together can spin the spacecraft with minimal residual ΔV. They are suitable for quick turns and get new orientations in few seconds. Redundancy is granted, in fact the RCS is still able to work in case of failure of one thruster in any paired couple, even if some lateral velocity is given [8]. The RCS is also used for the desaturation procedure of the RWs, reducing their accumulated angular momentum. In fact, since MRO is only

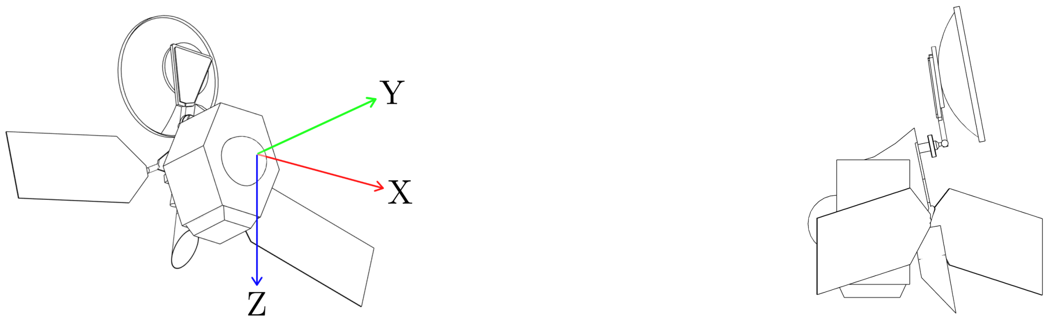
symmetric along one axis, a non-zero net torque is applied to the orbiter due to various disturbances nearly at all times, which is counteracted by RWs until they are saturated. In general, the RCS is employed whenever the RWs control system is not suitable or unavailable, such as safing events, trajectory correction maneuvers (TCMs), MOI and aerobraking [4]. Each thruster has a mass of 0.33 kg, a nominal thrust of 0.9 N. The specific impulse is estimated between 209 and 224 seconds, while the minimum impulse bit stays around 0.27 Ns [1]. While the placement of thrusters are not publicly available, they are presumed paired and equidistant from the shifting axis of the Center of Mass (CoM).

1.2.3 Flight Control Software

The attitude estimation is performed by a 6-state extended Kalman filter. The on-board computer combines the quaternion estimates from the star trackers with angular speed measurements from MIMU to produce a 3-state error estimate and a 3-state gyro bias estimate. The uplink system provides ephemerides and pointing commands to the spacecraft, which processes and passes them on to the other components of the AOCS as attitude commands [5].

1.3 Pointing Budget and Modes Correlation

The spacecraft body-fixed frame is defined such that +Z is aligned with the normal direction of the nadir deck, +Y is oriented along the centerline of the propellant tank and the MOI thrusters, and +X is defined by the cross product of +Y and +Z, as shown in Figure 1a. The origin of the frame is located at the geometric center of the launch separation adapter plate. Both the HGA and solar arrays are equipped with gimbals able to rotate along their inner and outer rotation axes. Different attitude configurations are adopted throughout the mission, depending on the specific requirements of each phase and operational mode. The following paragraphs describe the main attitude configurations during different modes of the main mission phases.



(a) (b)

Figure 1: **(a)** Simplified model of MRO. **(b)** MRO as seen from the drag direction in PSP.

Checkout Mode (CM): right after the separation from the launcher, the orbiter was released with an angular velocity of 0.3°/s around X axis. After a series of appendage-deployment activities, MRO was settled in a preset inertial-fixed acquisition attitude. In this configuration the -Y axis is pointed to the Sun with a bias error of 15° around the X axis, which is obtained by the cross product of Sun and velocity direction vectors. The solar arrays gimbal angles were null, while the inner and outer rotation angles of the HGA gimbal are respectively set to 180° and -45° [11].

Sun-point Cruise Mode (SCM): about three days after launch the spacecraft transitioned into this configuration, pointing the -Y axis to the Sun. The -X axis is obtained by the cross product between the Sun and Earth direction vectors. The solar gimbal angles are null, the inner angle of the of the HGA is fixed at 180°, while the outer angle is the result of the Sun-probe-Earth angle minus 90°. This configuration ensures that the Earth is always in the YZ plane and allows the HGA to easily track the Earth maintaining its gimbal angle constant [11].

Spread-Eagle Mode (SEM): two months before MOI, the orbiter assumed this attitude, with -Y axis pointed toward Earth and +Z axis result of the cross product between Earth and Sun direction vectors. Both solar arrays gimbal angles are set to zero, while HGA gimbal inner and outer angles are respectively set to 180° and -90°. This configuration allows to maximize the cross area of the HGA and exploit air drag during MOI. This attitude is also used for TCMs and AMD events [11], exploiting MRO's asymmetry to aid in attitude control.

Aerobraking Mode (AM): this configuration is analogous to the previously described spread-eagle mode, except with lower pointing accuracy due to harsh conditions.

Global Mapping Mode (GMM): this is one of the three observation modes assumed during the Primary Science Phase (PSP). The instruments involved, MCS and MARCI, require nadir pointing, low data rate and continuous operations. This means that the +Z direction is aligned with the nadir one, while the HGA and solar arrays gimbal angles vary to track respectively the Earth and the Sun.

Survey Mode (SUR): this configuration is assumed to allow CRISM, CTX and SHARAD to work properly. The attitude is similar to GMM, but requires much more precise pointing accuracy.

Target Investigations Mode (TIM): this observation mode includes the usage of the CTX, the CRISM and HiRISE. The target investigations often requires off-nadir rolls up to 30° for HGA and solar arrays contraints. However, rolls over 9° are limited to one per day due to the effects on MCS studies. The nominal attitude is still the same as the other observation modes (+Z axis aligned with nadir direction), but the HiRISE requires incredibly high pointing accuracies as reported in Table 1.

Safe Mode (SAFE): if an anomaly occurs, MRO switches to Safe Mode, a stable configuration. In this mode, LGA1 is pointed toward Earth, and the solar arrays are Sun-pointed, with the spacecraft –Y axis tracking the Sun. Sun and Earth ephemerides loaded before launch are used to determine the Sun–probe–Earth angle and guide the initial pointing of the Low-Gain Antenna (LGA). Star trackers can be used for Sun acquisition. If star trackers are unavailable, and the only attitude knowledge comes from Sun sensors, the spacecraft rotates about the –Y axis, causing the LGA boresight to trace a cone relative to Earth, thus signaling the presence of an anomaly to the ground stations [10].

Mode	**Main Tasks**	**APE [mrad]**
Checkout	Detumbling, appendage-deployment, initial attiude determination	-
Sun-Point Cruise	HGA communications, solar arrays Sun-pointing, calibrations	2.08 (HGA)
Spread-Eagle	MOI preparation, wheels desaturation, air drag and asymmetry exploitation	-
Aerobraking	Air drag exploitation	-
Global Mapping	MARCI and MCS continuous operations	3.0 (MCS)
Survey	CRISM, CTX and SHARAD extended nadir observations	0.020 (CTX)
Target Obs	CTX and CRISM operations, HiRISE high precision target observations, roll maneuvers	0.0032 (HiRISE)
Safe Mode	LGA communications, solar arrays Sun-pointing	-

Table 2: Control Modes and Pointing Budgets

1.4 Reverse Sizing of the AOCS

In this section, the attitude control system of the MRO is analyzed starting from mission requirements and performance objectives, considering how sensors, actuators, and environmental disturbances interact. Through a reverse sizing approach, each component is iteratively studied to ensure it can reasonably meet operational needs across all mission phases. This process justifies design choices in terms of sizing, redundancy, and allocation of key resources such as mass, power, and data.

1.4.1 Mass and Inertia Properties

Using the simplified model, with remaining propellant mass corresponding to that of the PSP initialization, the following physical properties are obtained. While they are varying in reality, these values are held constant across calculations for different phases.

Parameter	Value	Unit
Mass	1897	kg
$I_x = I_{\max}$	2898	$\mathrm{kg\,m^2}$
$I_y = I_{\min}$	1961	$\mathrm{kg\,m^2}$

Table 3: Physical properties relevant to AOCS sizing.

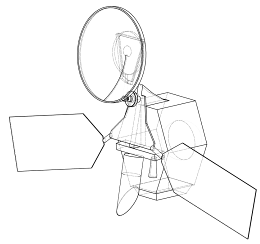


Figure 2: The simplified MRO model.

1.4.2 External disturbances

During the mission, the interaction with external disturbances affected the attitude dynamics of the spacecraft. The main ones are the following: gravity gradient, atmospheric drag and solar radiation pressure (SRP). The torques, resulting from these disturbances, are evaluated with the following equations:

Disturbance	**Reference formula**	**Result** [N m]
Gravity gradient	$T_{\mathrm{GG}} = \dfrac{3\mu_{\mathrm{Mars}}}{2R_{\mathrm{orb}}^3}(I_{\max} - I_{\min})\sin(2\theta)$	1.445×10^{-3}
Solar radiation pressure	$T_{\mathrm{SRP}} = \dfrac{F_S}{c} A_{\mathrm{Sun}}(1+q)\cos(\mathrm{i})\ \lVert\vec{r}_{\text{GC-CM}} \times \hat{n}_{\mathrm{SA}}\rVert$	9.08×10^{-4}
Aerodynamic drag	$T_{\mathrm{d}} = \dfrac{1}{2}\rho C_d A V^2 (c_p - c_g)$	5.7×10^{-3}

Table 4: Disturbance torques using MRO and Mars environment parameters.

For all the calculations, the worst case scenario is assumed, so the angle θ in the gravity gradient formula is considered the maximum off-nadir pointing angle of the HiRISE (30°). The incident angle in the expression of the SRP is assumed 0°, while the relative velocity V, used for the drag torque is assumed equal to the average orbital velocity of the spacecraft, 3405 km/s. For the majority of PSP, it is reported that the +X side of MRO is exposed to the drag [11]. The worst case, i.e. the maximum exposed area, is driven by the keep-out-zones of the gimbals [9], replicated using the simplified model as shown in Figure 1b.

1.4.3 Reaction Wheels

The four RWs are disposed, inside the spacecraft, in a configuration with three rotors aligned with three orthogonal axis and the fourth with equal components along these axis. This configuration can be expressed in matrix form with each row having a unitary versor in its corresponding axis, and $1/\sqrt{3}$ throughout the fourth column, representing the fourth wheel.
Attitude station keeping: starting from the total torque applied on the spacecraft, the torque of each single reaction wheel, to counteract external disturbances and keep the MRO aligned, is evaluated considering the worst case, with external torque concentrated along one of the three orthogonal axis: $T_{\text{RW\_disturbance}} = \frac{T_d}{1+\frac{1}{\sqrt{3}}} = 4.6 \times 10^{-3}\text{N m}$ (for PSP case). Under the influence of external disturbances, the angular momentum accumulated by a single reaction wheel results: $h_{\text{RW}} = T_{\text{D}} \cdot t_{\text{orb}}$, so the period of time after which it is necessary a desaturation maneuver can be estimated, and is evaluated in the next section.
Slew maneuver: during the mission the ADCS subsystem must be able to perform slew maneuvers and realign the pointing direction with the target one. The slew maneuver with RWs, is performed with a constant acceleration and breaking. Considering the maximum torque, the slew rate go above the maximum value imposed ($\dot{\theta}_{\text{slew,max}} = 0.5°/\text{s}$), so a new torque for the maneuver is computed, $T_{slew} = 0.16\text{N m}$, lower than the maximum, so the time required for the slew maneuver:

$$t_{\text{slew,RW}} = \sqrt{\frac{4 \cdot \theta_{\text{slew,max}} \cdot I_{\text{max}}}{T_{\text{slew}}}} = 466.10s$$

The maximum angular momentum, reached during the slew maneuver, results:
$h_{\text{slew,max}} = T_{\text{max}} \cdot t_{\text{slew,min}} = 25.29\text{N m s}$, lower than the maximum angular momentum that can be stored by these RWs. Then the mechanical power provided by the RW is calculated and converted in the electrical power requested using their electric-to-mechanical efficiency [3].

1.4.4 Thrusters

The configuration of thrusters is characterized by four banks, each with two thrusters. The reverse sizing is performed considering that only four thrusters contribute to the total torque and they are disposed symmetrically with respect to the center of mass. The sizing of thrusters is realized considering three main attitude control scenarios:
Attitude station keeping: the necessary torque to counteract the effects of external disturbances and maintain the desired pointing, result from the following equation: $F_{\text{th}} = \frac{T_D}{nL} = 9.08 \times 10^{-4}\text{N m}$ (for PSP case).
Slew maneuver: with thrusters to get a desired slew angle, can be achieved in different maneuvering sequences: acceleration, coasting and breaking. The maximum thrust F_{th} is equal to 0.9 N, taken from datasheet, the resulting torque applied is: $T = nF_{\text{th}}L$, the time of the maneuver is given by:

$$t_{\text{acc}} = t_{\text{brake}} = \frac{\dot{\theta}_{\text{acc,max}} I_{\text{max}}}{nFL} \qquad t_{\text{coast}} = \left(\theta_{\text{slew,max}} - \frac{nF_{\text{th}}L}{I_{\text{max}}} t_{\text{acc}}^2\right) \cdot \frac{1}{\dot{\theta}_{\text{coast}}}$$

with $t_{\text{slew}} = t_{\text{acc}} + t_{\text{coast}} + t_{\text{brake}} = 363.51\text{s}$. The propellant mass for a single slew maneuver can be evaluated with the specific impulse I_{sp} and the total impulse I_{tot}: $M_{\text{prop}} = \frac{I_{\text{tot}}}{I_{\text{sp}}} = 0.0031\text{kg}$, with $I_{\text{tot}} = t_{\text{slew}} \cdot F_{\text{th}}$.
Desaturation: considering data of thrusters and RWs, from the previous section, the time required to desaturate using the maximum thrust is higher then typical value reported in literature, so the thrust to desaturate is computed as follow: $F_{\text{desat}} = \frac{h_{\text{max}}}{nLt_{\text{desat}}} = 0.4167\text{N}$. The number of desaturation events can be evaluated computing the period before reaching saturation and the duration of the phase of interest, the propellant mass required can be estimated using the aforementioned formula but with a total impulse given by: $I_{\text{tot}} = n \cdot t_{\text{desat}} F_{\text{desat}}$.

1.4.5 Mode Correlations

In the following tables, the correlation between the operational modes and the most relevant AOCS parameters and technologies is presented. Only the dominant disturbance sources are listed, and for sensors and actuators, only the primary devices involved in each mode are highlighted.

Phase	**Mode**	**Dist.**	**Act.**	**Sens.**	**Slew Sizing**	**Desaturation Sizing**
LEOP	CM	N/A	RW	Sun S., MIMU	N/A (Detumbling)	N/A
Cruise	SCM	SRP	RW	Sun S., MIMU	$n_{\text{slew,day}} = 6$ $E_{\text{day}} = 46.41\text{W h}$	$n_{\text{des,day}} < 1$ $M_{\text{prop,tot}} = 0.1\text{kg}$
	SEM	SRP	RW, RCS	MIMU, Star T.		
ATP	AM	Drag, SRP	RW, RCS	MIMU, Star T.	$n_{\text{slew,day}} = 29$ $E_{\text{day}} = 258.41\text{W h}$	$n_{\text{des,day}} = 5$ $M_{\text{prop,tot}} = 7.08\text{kg}$
PSP	GMM	Drag	RW	Star T., MIMU	$n_{\text{slew,day}} = 31$ $E_{\text{day}} = 276.96\text{W h}$	$n_{\text{des,day}} = 6$ $M_{\text{prop,tot}} = 26.42\text{kg}$
	SUR	SRP	RW			
	TIM	Grav.	RW, RCS			

Table 5: AOCS Parameters by Phase and Mode

In Table 5, the slew maneuvers are considered to be performed exclusively using RW, while desaturation is achieved solely by using the RCS thrusters. For completeness, a propellant consumption of 0.0031 kg is designated for a full slew maneuver relying only on thrusters. It should be noted that the $M_{\text{prop,tot}}$ listed in the table refers to the total propellant mass consumed during all of considered phases only to desaturate the S/C.

1.4.6 Mass, Power and Data Budgets

Component	**Mass** [kg]	**Power** [W]	**Data Rate** [kbps]
Reaction wheel assembly (RWA)	48 (12 × 4)	68.53	~ 0
RCS	32.74	-	~ 0
IMUs	9.2 (4.6 × 2)	25	23.04
Star trackers	7.2 (3.6 × 2)	8.9	2.56
Sun trackers	3.4 (0.215 × 16)	0 (Passive)	~ 0
Total	101	102	~26 (~31 with margin)

Table 6: Mass, power, and data budgets of the AOCS, including redundancy.

The data rates are calculated using values (during slew) found in similar spacecraft, and supporting data found in the literature, such as float precisions and refresh rates. It is assumed that sensor outputs are 3+3 state matrices and attitude quaternions. A margin of 20% is applied to the total data rate to account for housekeeping data. The power for the RWA, reported in the table above, is computed to realize a slew maneuver of 180° with a single RW.

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Acronyms

AM Aerobraking Mode. 4, 7

AMD angular momentum desaturation. 4

AOCS Attitude and Orbit Control Subsystem. 1, 3, 5, 7

APE Absolute Performance Error. 4

CCD Charge-Coupled Device. 2

CM Checkout Mode. 3, 7

CoM Center of Mass. 3

CRISM Compact Reconnaissance Imaging Spectrometer for Mars. 1, 4

CTX Context Camera. 1, 4

DC Direct Current. 2

FIT Failure in Time. 2

FOV Field-of-view. 1, 2

GMM Global Mapping Mode. 4, 7

HGA High-Gain Antenna. 1, 3, 4

HiRISE HIgh Resolution Imaging Science Experiment. 1, 4, 5

IMU Inertial Measurement Unit. 7

LGA Low-Gain Antenna. 4

LGA1 Low-gain Antenna 1. 4

MARCI Mars Color Imager. 4

MCS Mars Climate Sounder. 1, 4

MIMU Miniature Inertial Measurement Unit. 2, 3, 7

MOI Mars Orbit Insertion. 3, 4

MRO Mars Reconnaissance Orbiter. 1–5

NEA Noise Equivalent Error. 2

ONC Optical Navigation Camera. 1

PSP Primary Science Phase. 3–6

RCS Reaction Control System. 1–3, 7

RW reaction wheels. 1–3, 6, 7

RWA Reaction wheel assembly. 7

S/C spacecraft. 7

SAFE Safe Mode. 4

SCM Sun-point Cruise Mode. 3, 7

SEM Spread-Eagle Mode. 4, 7

SHARAD SHAllow Subsurface RADar. 4

SRP solar radiation pressure. 5

SUR Survey Mode. 4, 7

TCM trajectory correction maneuver. 3, 4

TIM Target Investigations Mode. 4, 7

TRL Technology Readiness Level. 2

TTMTC Tracking, Telemetry and Telecommand Subsystem. 1